

Practical Experiment: Compare the response speed of open-source Large Language Models (LLMs) to find the fastest model for real-time applications

Meeting 2: Comparative Analysis & Measure Factual Accuracy in LLM Responses

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Agenda

- Introduction & Architecture Overview
 - System Graph & Data Flow
 - Component Specifications
 - Natural Language Toolkit (NLTK) Data Management
 - Natural Language Inferencing (NLI) Model Specifications
 - Bilingual Evaluation Understudy (BLEU) Score Matrix
 - Smoothing Methods Comparison
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Introduction & Architecture Overview

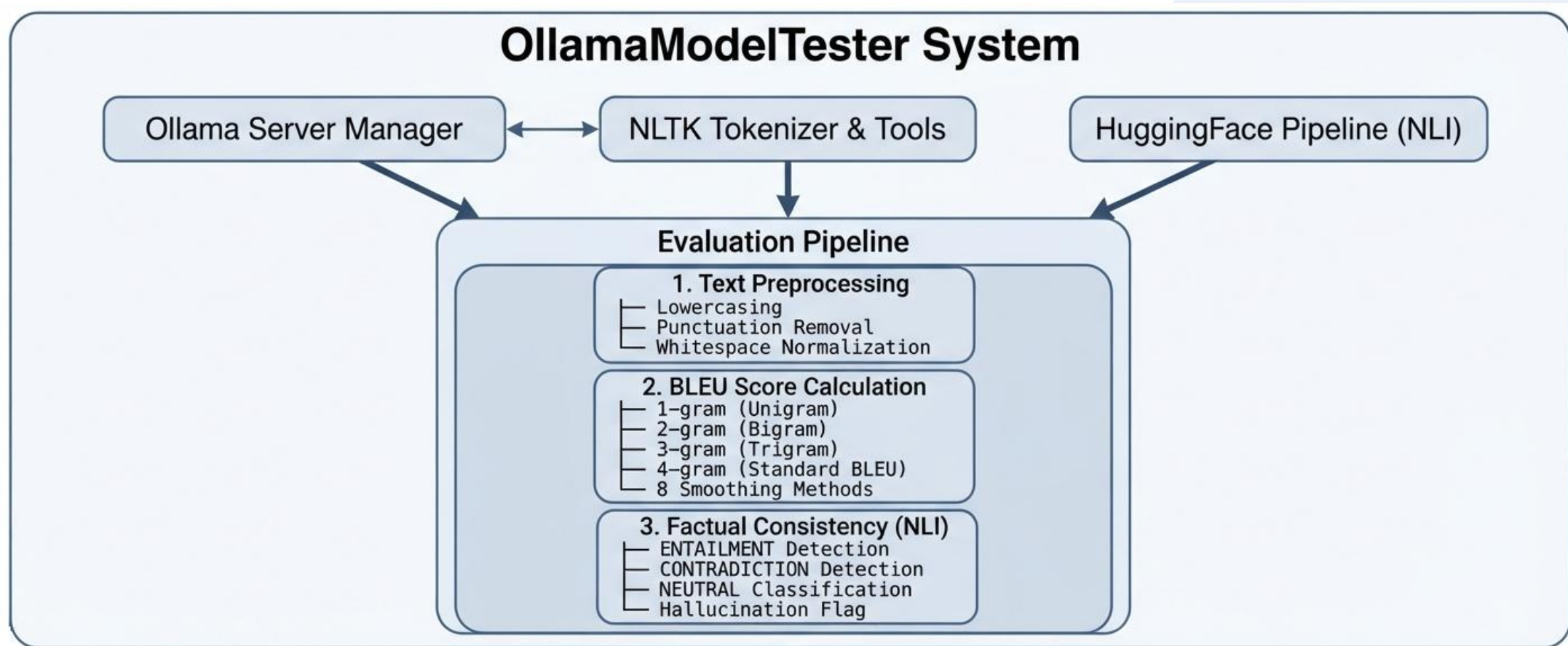
Objective: Present evaluation framework that pairs semantic verification (Natural Language Inference) with lexical overlap metrics (BLEU scores).

Core Components:

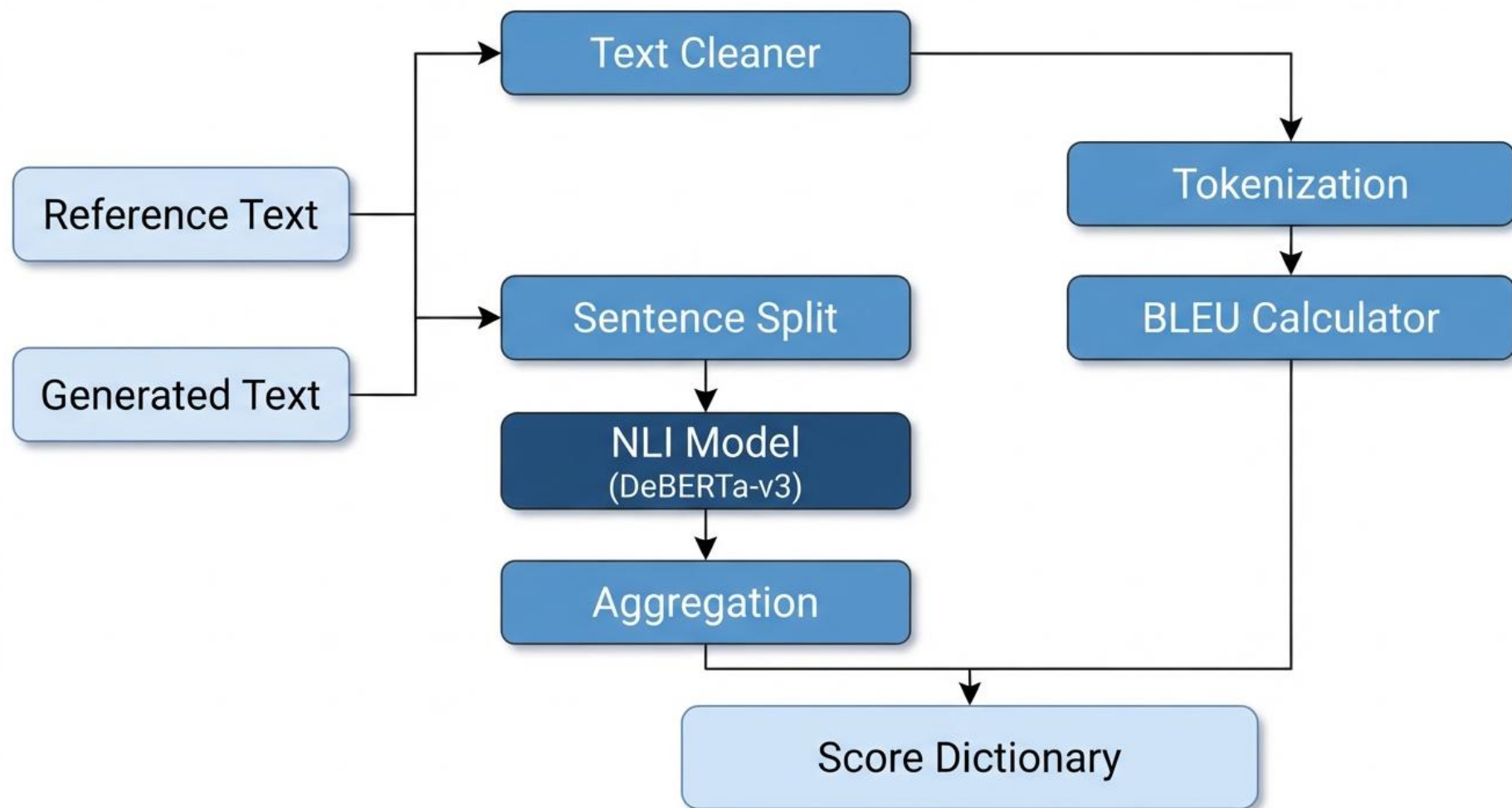
- **NLTK Integration:** Ensures linguistic tokenizers and taggers are available locally.
- **NLI Cross-Encoder:** Detects factual consistency, contradictions, and hallucinations at the sentence level.
- **BLEU Evaluator:** Computes multi-variant n-gram overlap metrics using comprehensive smoothing techniques.



Introduction & Architecture Overview



System Graph & Data Flow



Component Specifications

- **Natural Language Toolkit (NLTK) Data Management**

Component	Purpose	Dependencies
punkt_tab	Sentence tokenization	NLTK >= 3.8
punkt	Legacy sentence tokenizer	NLTK >= 3.0
averaged_perceptron_tagger_eng	Serve as pre-trained Part-of-Speech (POS) tagger for English text. It maps raw text tokens to their corresponding grammatical categories based on the context of the sentence	NLTK >= 3.9

Component Specifications

- Natural Language Inferencing (NLI) Model Specifications

Property	Value	Impact
Model Name	cross-encoder/nli-deberta-v3-base	NLI model
Parameters	184 million	Good balance of speed/accuracy
Training Data	MNLI (Multi-Genre NLI)	Diverse text genres
Outputs	ENTAILMENT, NEUTRAL, CONTRADICTION	3-class classification
Input Format	"{premise} </s> {hypothesis}"	Standard NLI format

Component Specifications

- Bilingual Evaluation Understudy (BLEU) Score Matrix

Metric	Formula	N-Gram Weight	Purpose
BLEU-1	Precision of 1-gram	(1.0, 0.0, 0.0, 0.0)	Word overlap
BLEU-2	Precision of 2-gram	(0.5, 0.5, 0.0, 0.0)	Phrase matching
BLEU-3	Precision of 3-gram	(0.33, 0.33, 0.33, 0.0)	Sentence structure
BLEU-4	Combined 1-4 grams	(0.25, 0.25, 0.25, 0.25)	Standard evaluation
BLEU-AVG	Average of all	N/A	Overall quality

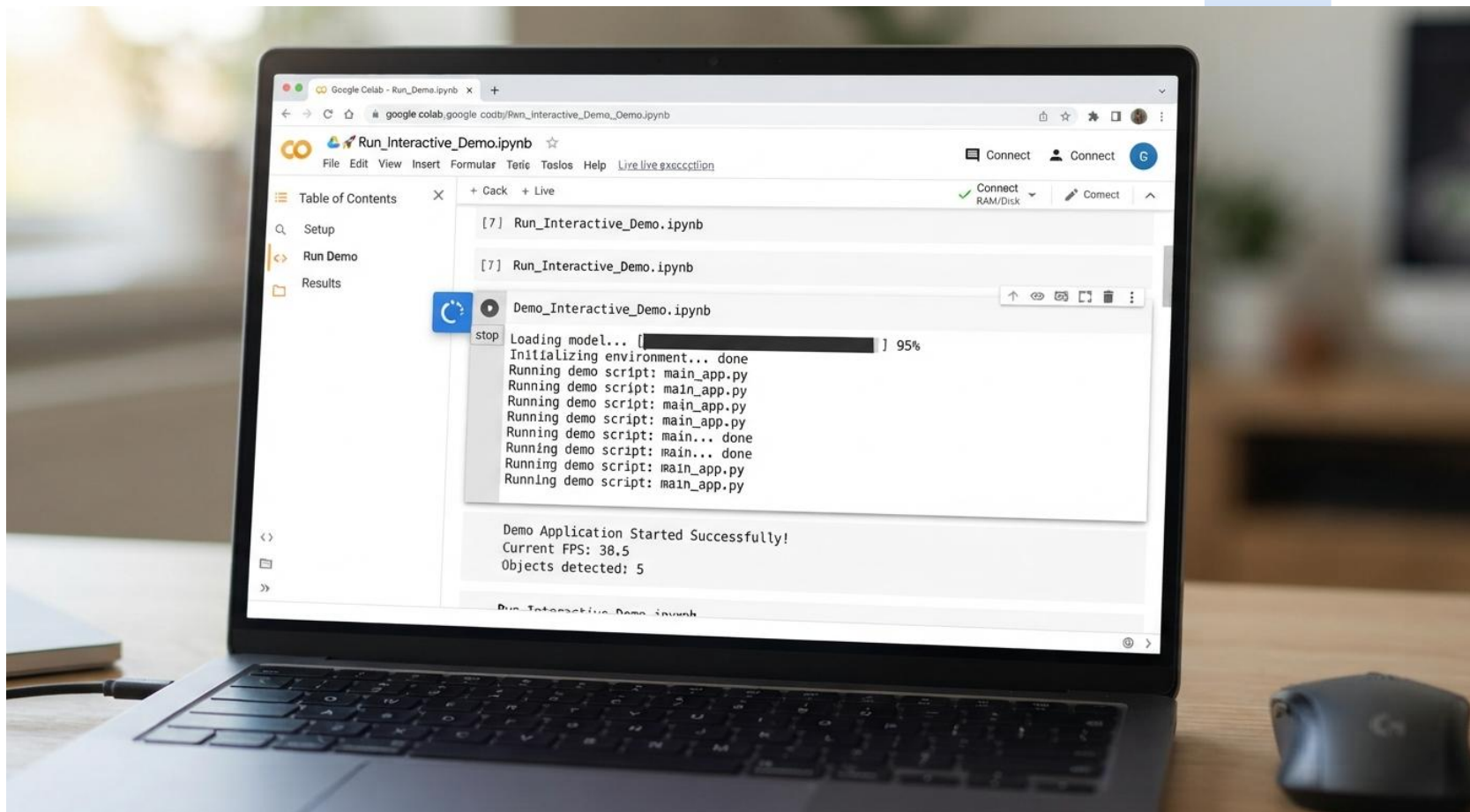
Component Specifications

- Smoothing Methods Comparison

Method	Best For
method0	Evaluating long texts.
method1	General purpose smoothing. It's a good default choice to avoid zero scores for short or non-overlapping texts.
method2	A more conservative alternative to method1. Minimal modification.
method3	Cases where you want to implement a smoothing approach inspired by the official National Institute of Standards and Technology (NIST) evaluation scripts.
method4	Recommended for general use. Designed for sentence-level evaluation and research contexts where you want to use a smoothing method benchmarked and evaluated for academic purposes.
method5	Statistical smoothing.
method6	Combination approach.
method7	The most advanced variant. Designed to handle cases where short texts are evaluated.

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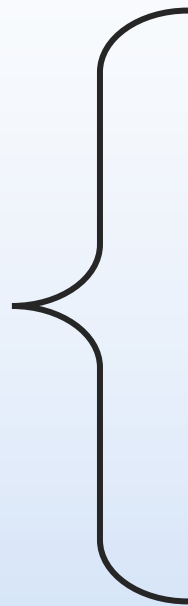
LIVE DEMO



References

- Colab – <https://colab.research.google.com/>
- Natural Language Processing (NLP) – <https://realpython.com/nltk-nlp-python/>
- Natural Language Processing (NLP) – <https://www.deeplearning.ai/resources/natural-language-processing>
- Natural Language Toolkit (NLTK) – <https://www.nltk.org/>
- Natural Language Inference (NLI) – https://sbert.net/examples/sentence_transformer/training/nli/README.html
- BLEU Score – <https://www.kaggle.com/code/mpwolke/bleu-score-bilingual-evaluation-understudy>
- BLEU Smoothing Methods – https://www.nltk.org/api/nltk.translate.bleu_score.html

Thank you



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